



Pulse Communication (PAM, PWM, PPM)

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Types of Pulse Communication

- Pulse Amplitude Modulation (PAM)

(Modulating Signal) *Analog*
or

- Pulse Width Modulation (PWM)

Digital

- Pulse Position Modulation (PPM)

Pulse Amplitude Modulation (PAM)

“It is the process in which amplitude of carrier pulse train is changed in accordance with the instantaneous amplitude of modulating signal keeping, width and position of carrier pulse train constant.”

PAM Waveforms

modulating

MESSAGE
SIGNAL
 $f_m = 1 \text{ kHz}$

multiplier $\times$

Carrier

SAMPLING
SIGNAL
 $f_s = 8 \text{ kHz}$

Pulses $\rightarrow$

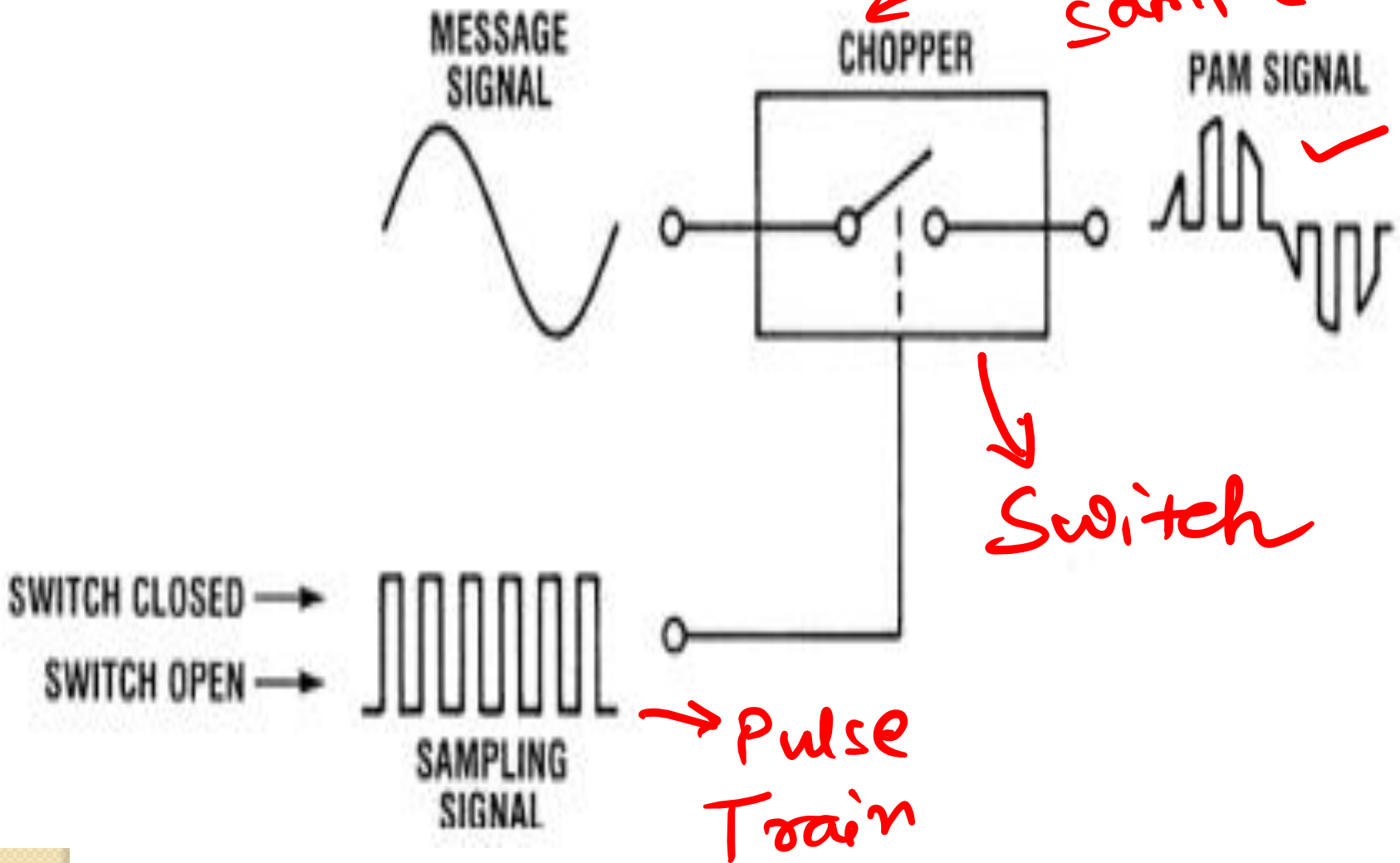
Sampled

Natural
PAM

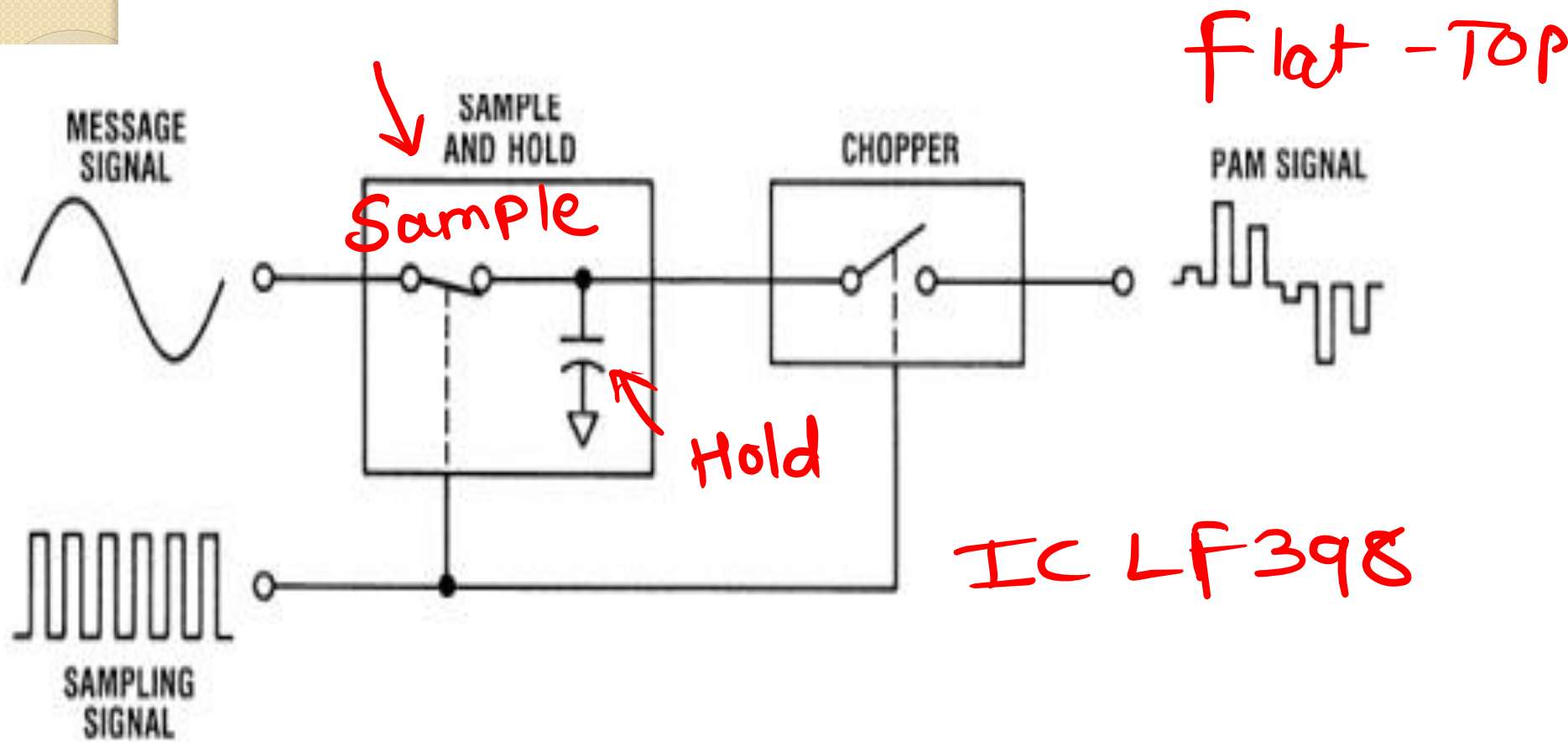
PAM
SIGNAL

Natural PAM

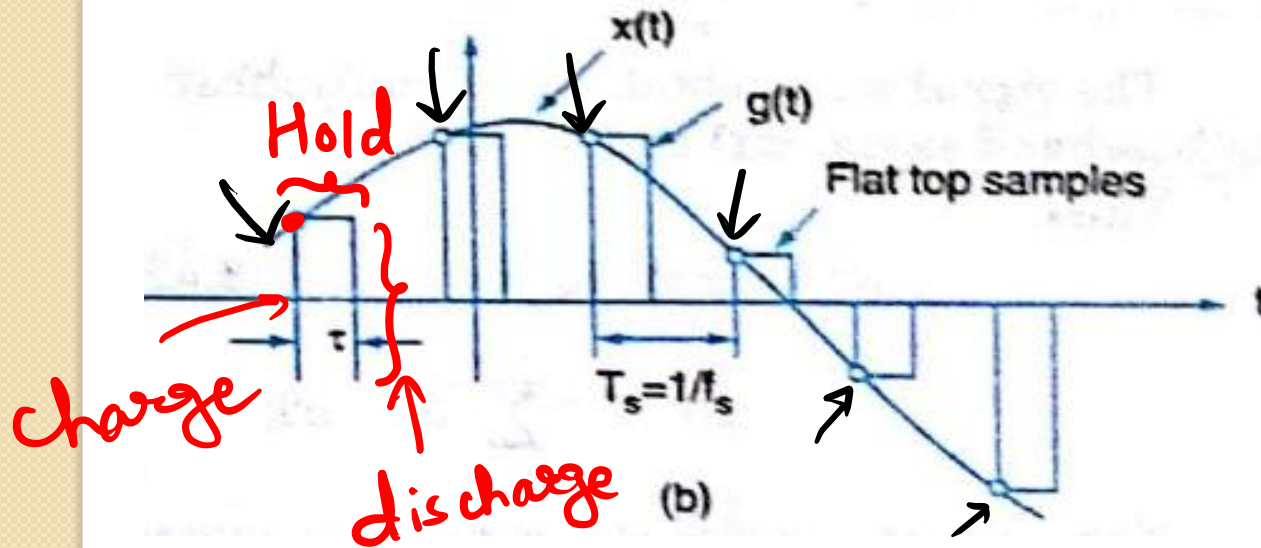
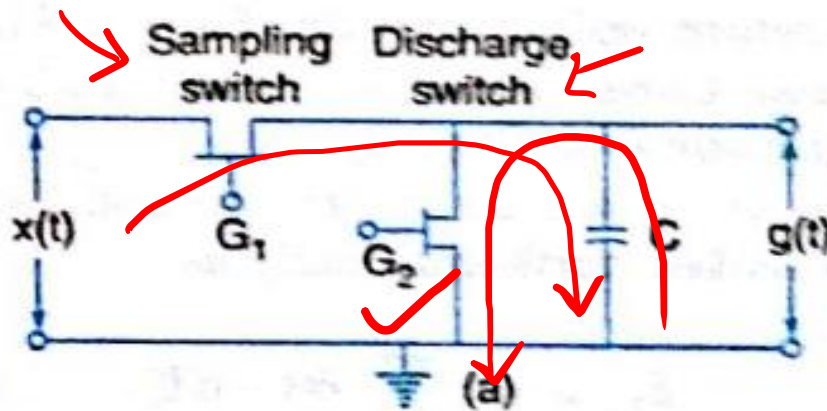
IC L398
sample hold



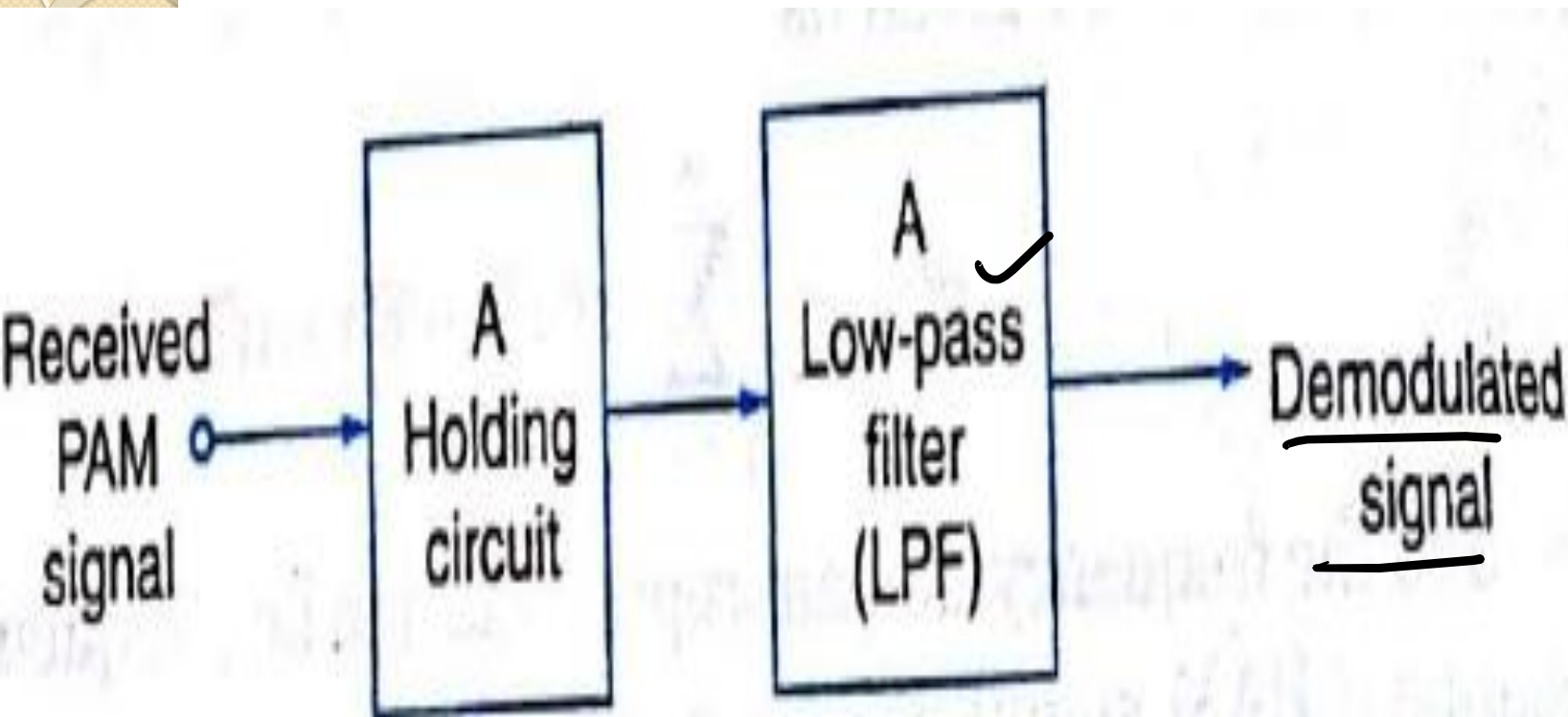
Flat Top PAM



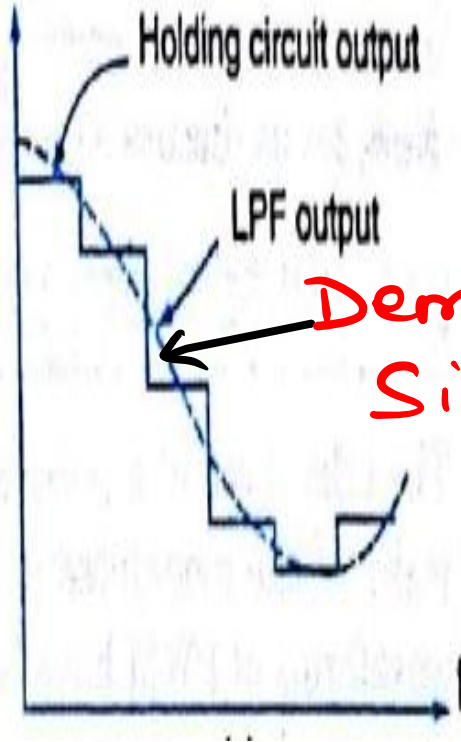
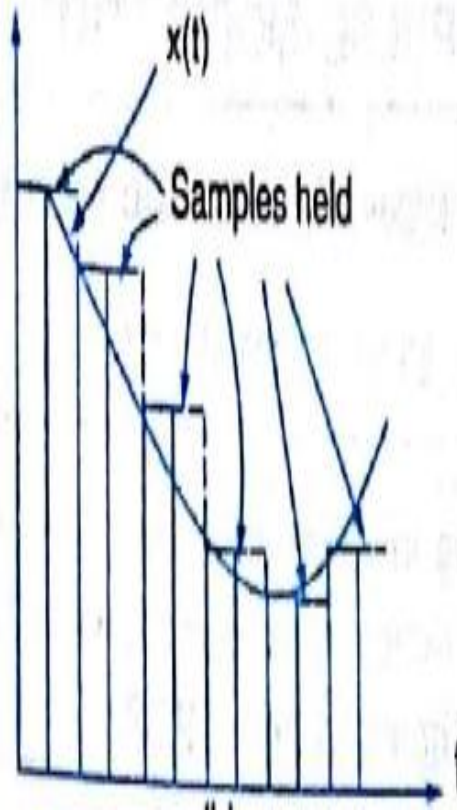
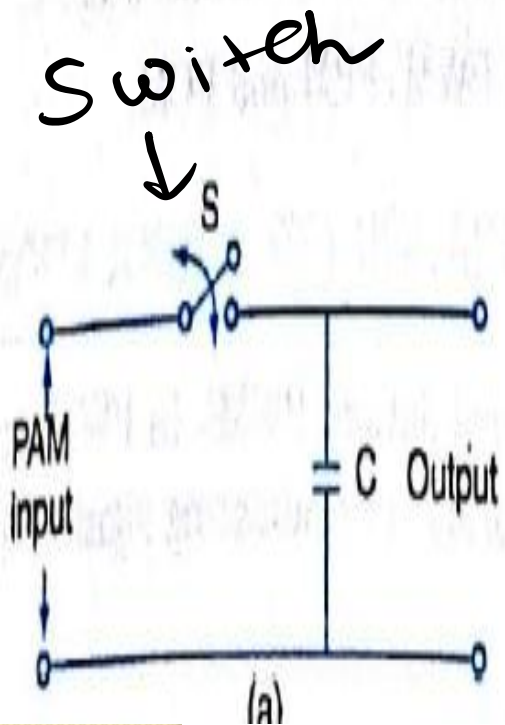
Flat Top PAM



PAM Demodulator Non-coherent



PAM Demodulator



Demodulated Signal

Drawback of PAM

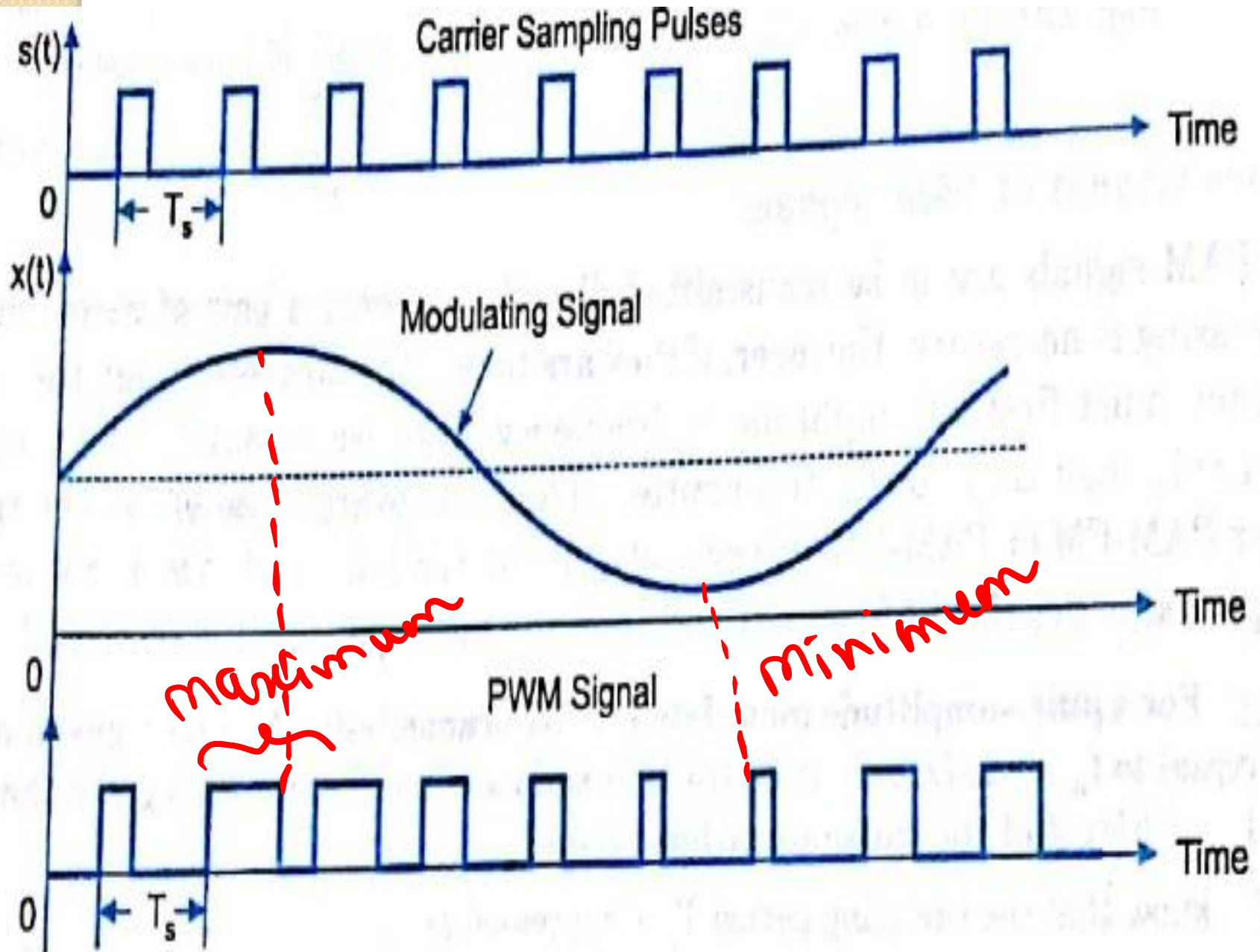
- Bandwidth requirement for PAM is higher as compared to other form.
- PAM signals have amplitude variations therefore interference of noise is maximum.
- Peak power required by the transmitter varies, since amplitude is varied in PAM.

Pulse Width Modulation (PWM)

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“It is the process in which width of carrier pulse train is changed in accordance with the instantaneous amplitude of modulating signal keeping, amplitude of carrier pulse train constant.”

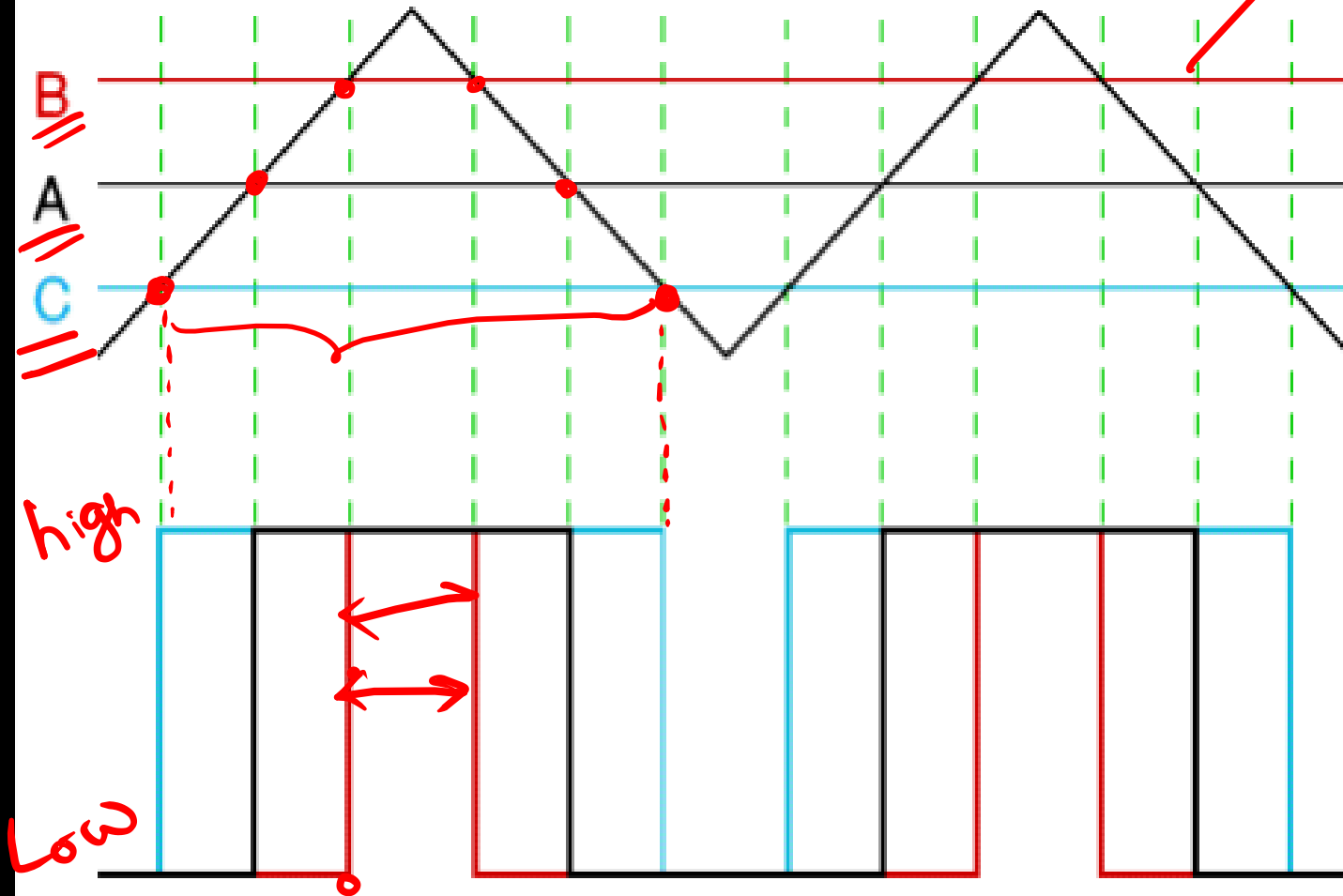
PWM Waveforms



Basic Principle of PWM

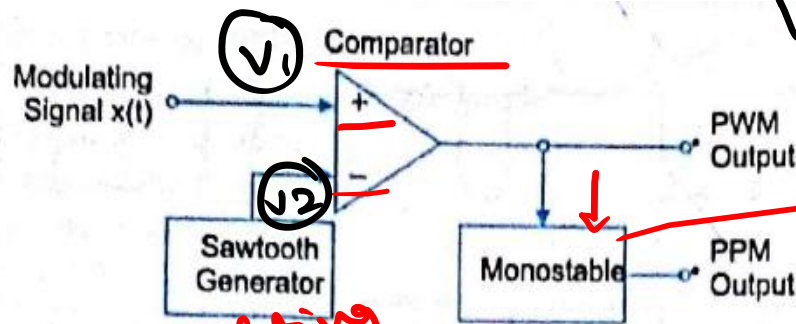
output goes low.

Comparison



IC 555
Timer
IC
↓
change
the
output
state

PWM and PPM Generation



output
 $V_1 > V_2 \rightarrow +ve$
 $V_1 < V_2 \rightarrow -ve$

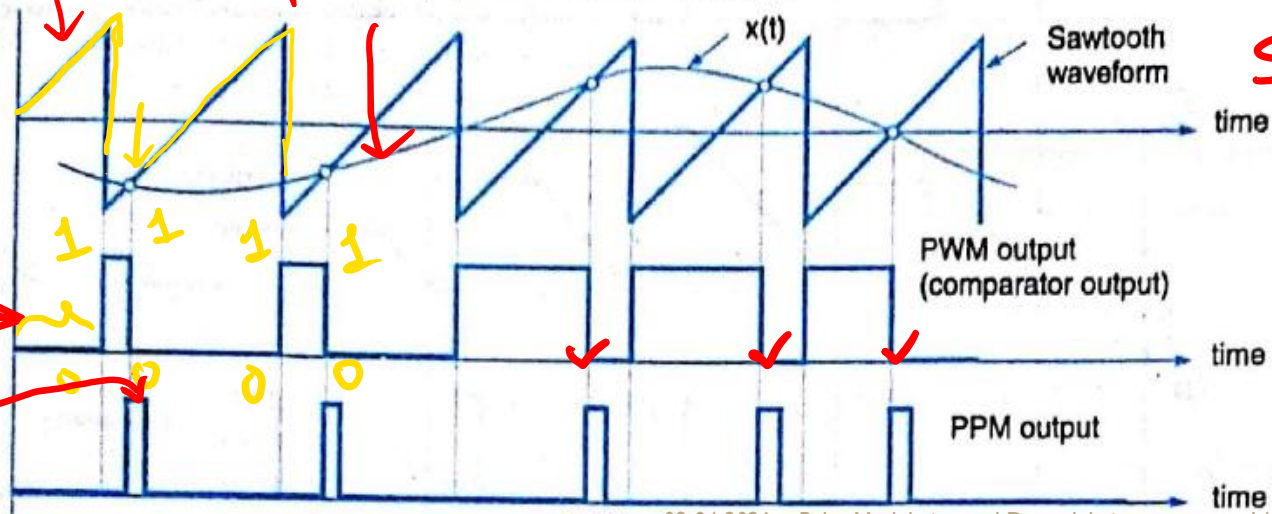
monostable
 multivibrator
 → Single State

State (stable)

Carrier

Modulating

(a) PWM and PPM Generator



$T_P = 1.1RC$

(b) Waveforms

09-04-2026

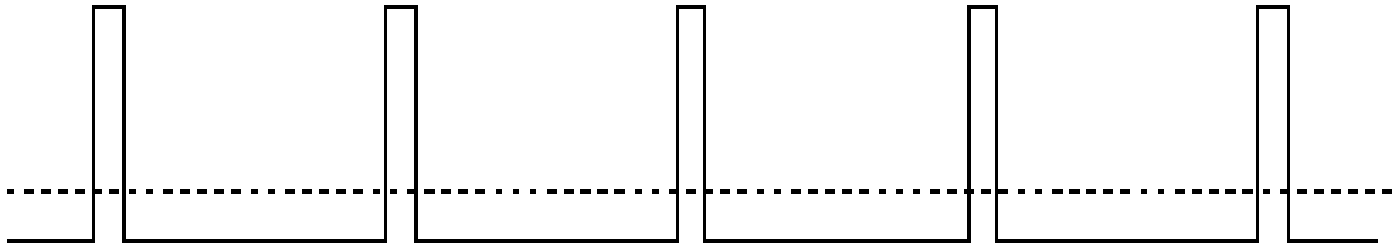
Pulse Modulation and Demodulation

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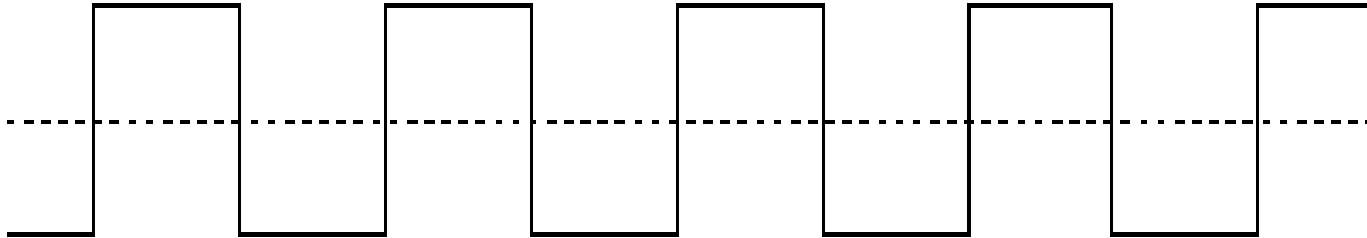
Waveforms of PWM with duty cycle variation

$$\frac{T_{ON}}{T_{ON} + T_{OFF}}$$

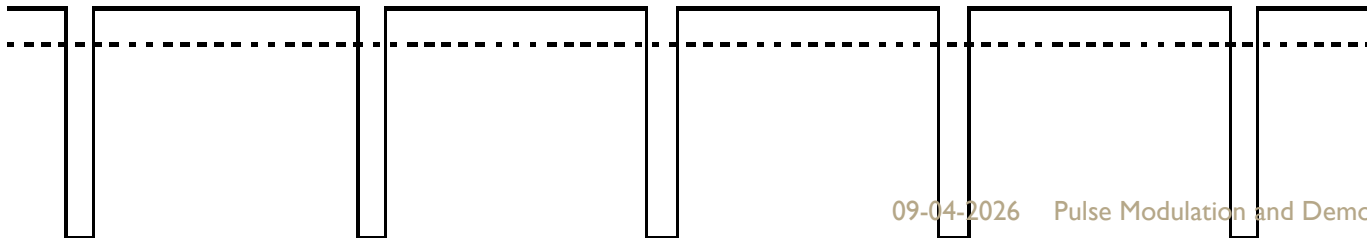
10% Duty Cycle



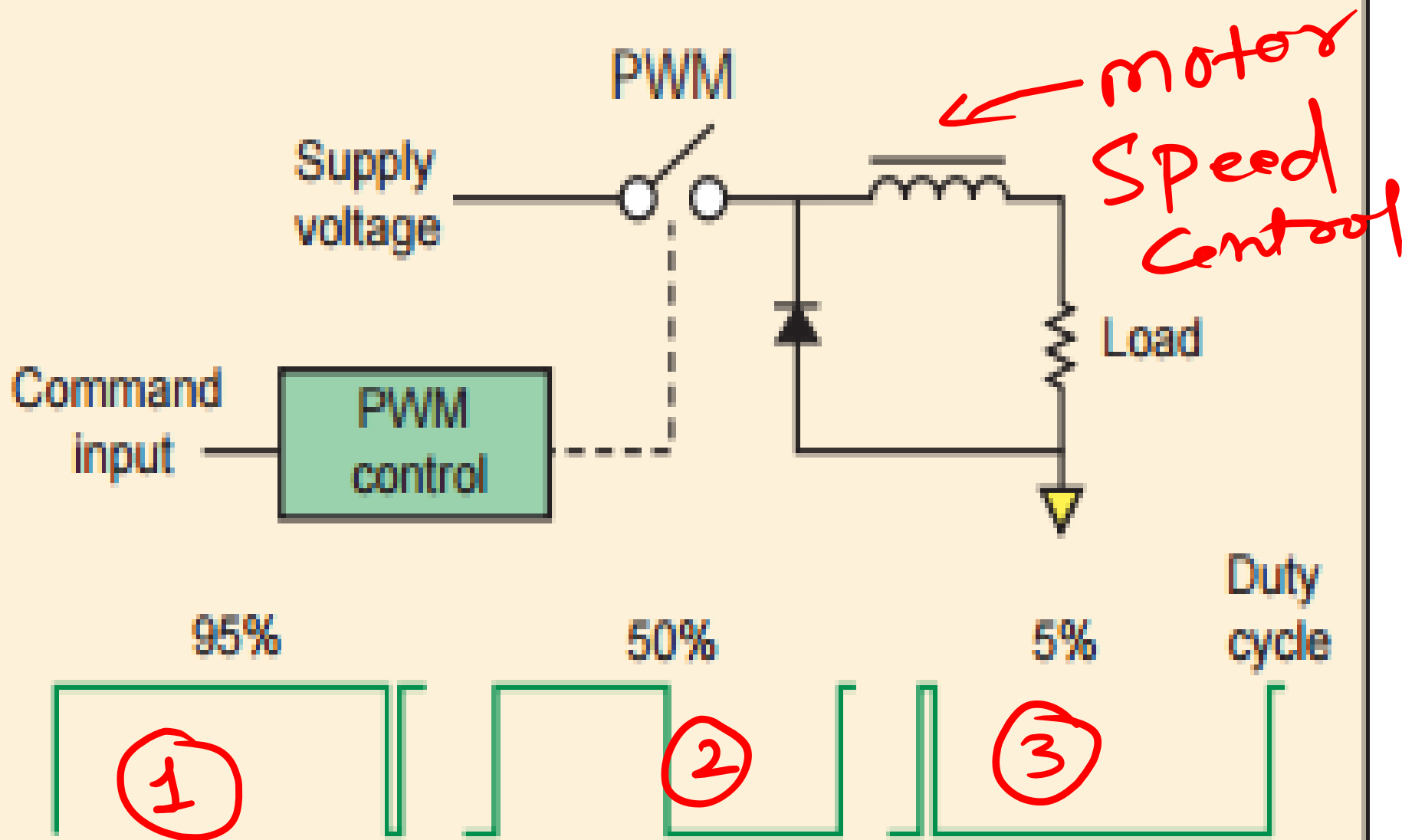
50% Duty Cycle



90% Duty Cycle



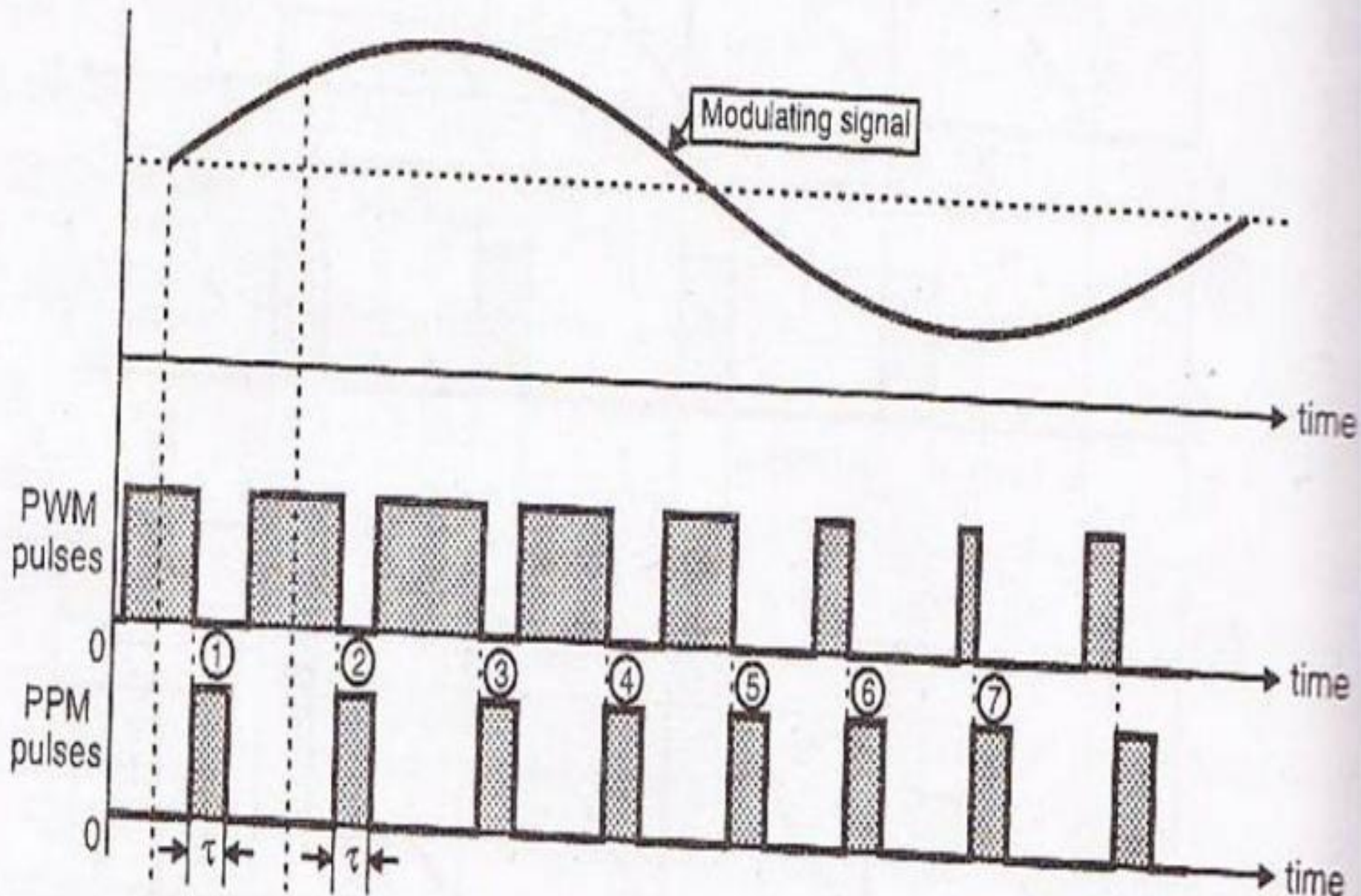
Application of PWM



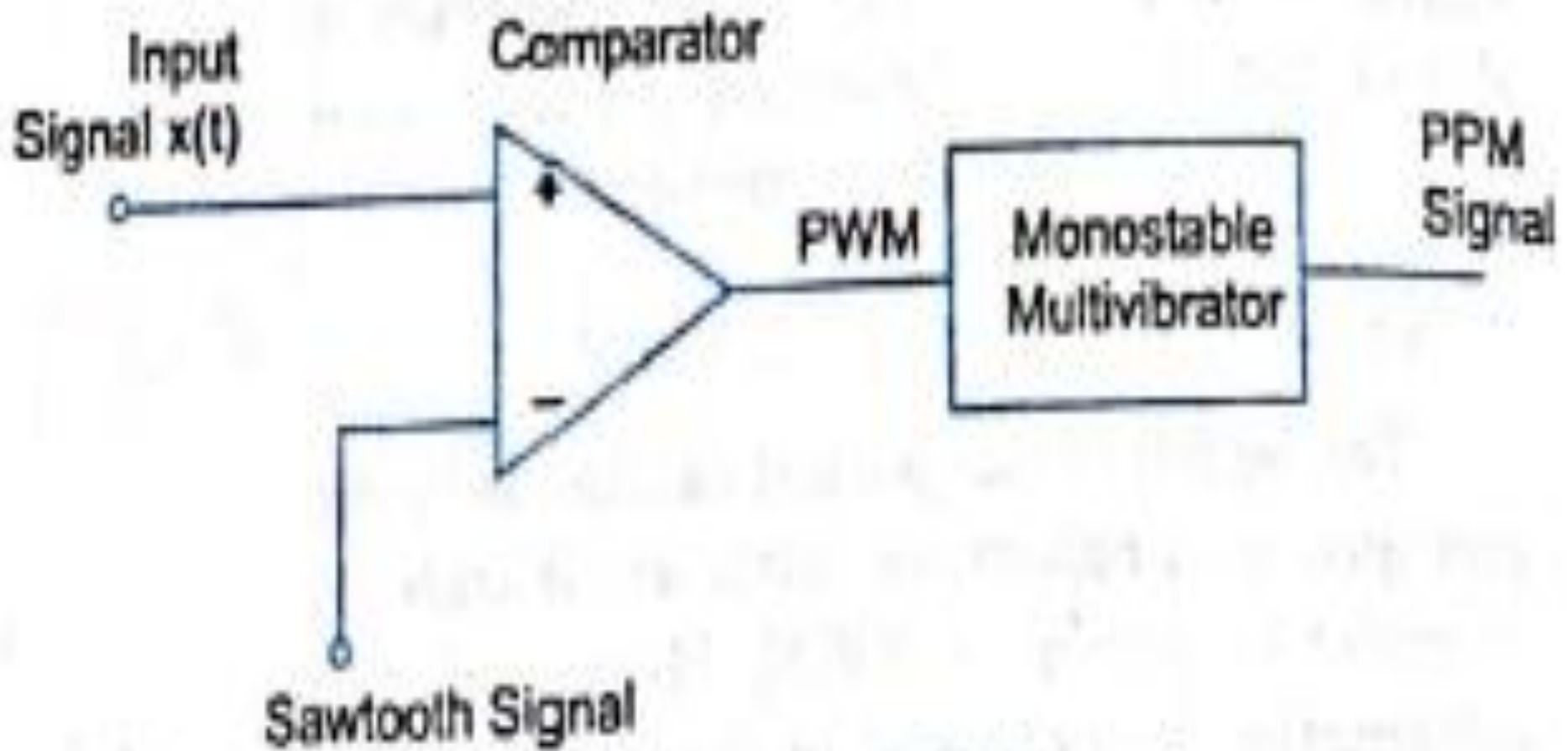
Pulse Position Modulation (PPM)

“It is the process in which position of carrier pulse train is changed in accordance with the instantaneous amplitude of modulating signal keeping, amplitude and width of carrier pulse train constant.”

PPM Waveforms

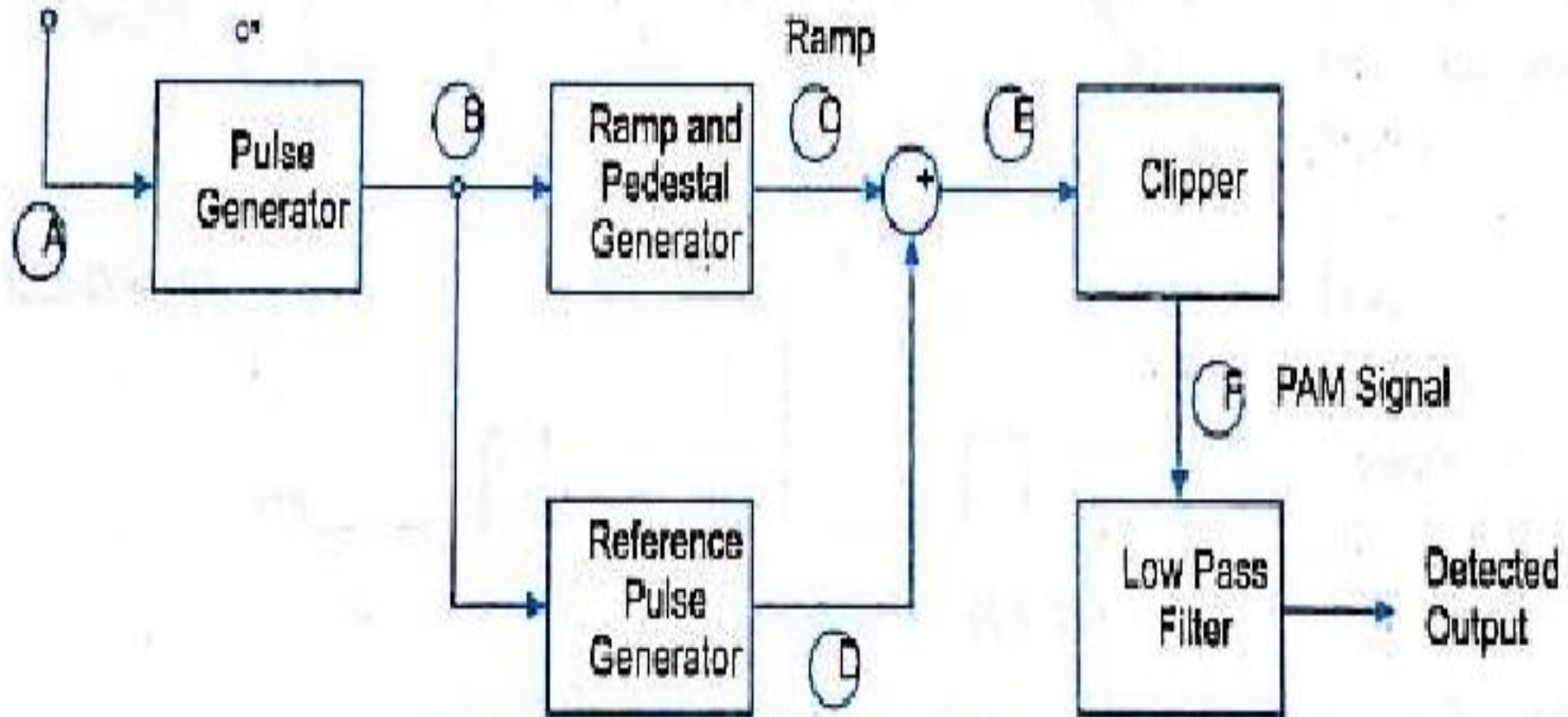


PPM Generation

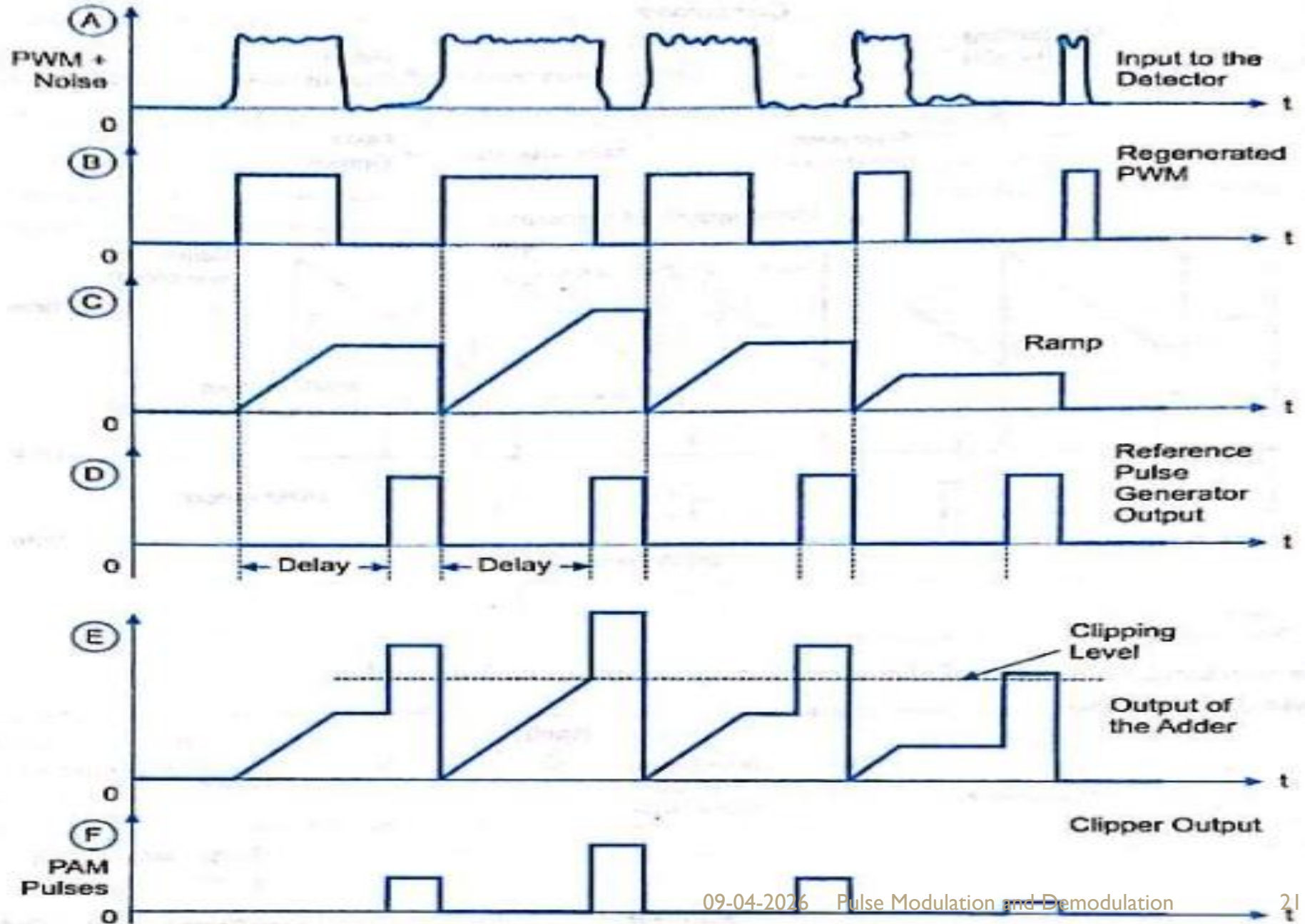


PWM Detection

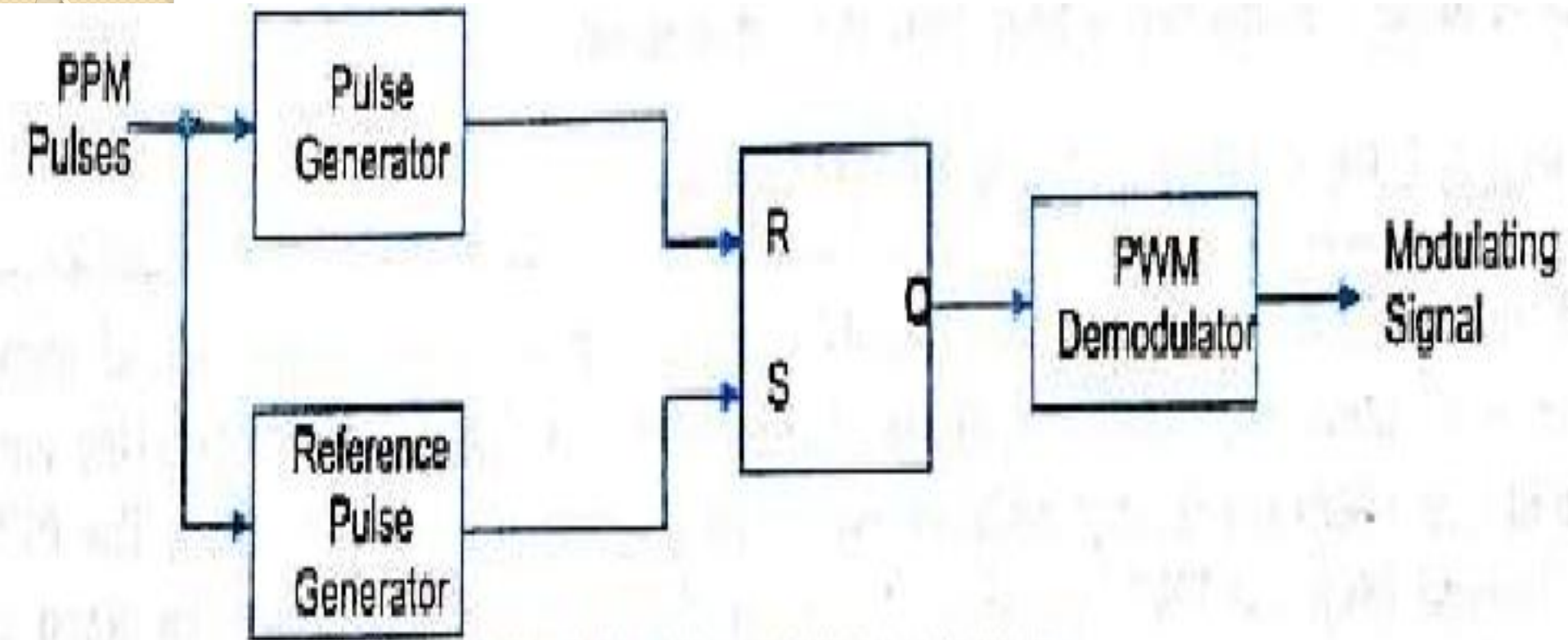
PWM Signal + Noise



PWM Detection

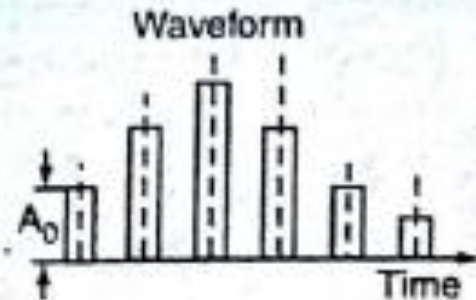


PPM Detection



Comparison of PAM/PWM/PPM

Pulse Amplitude Modulation (PAM)



Amplitude of the pulse is proportional to amplitude of modulating signal.

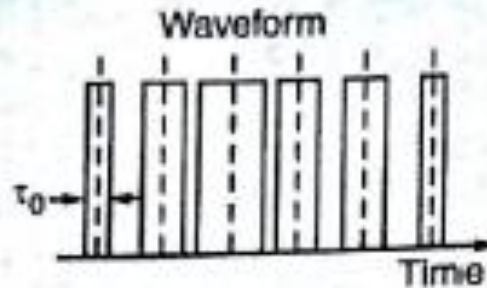
The bandwidth of the transmission channel depends on width of the pulse.

The instantaneous power of the transmitter varies.

Noise interference is high.
System is complex

Similar to amplitude modulation.

Pulse Width/Duration Modulation (PWM) or PDM



Width of the pulse is proportional to amplitude of modulating signal.

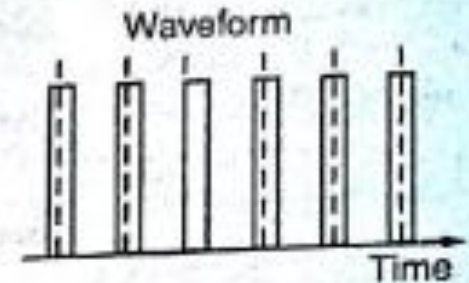
Bandwidth of transmission channel depends on rise time of the pulse.

The instantaneous power of the transmitter varies.

Noise, interference is minimum.

Simple to implement similar to frequency modulation.

Pulse Position Modulation (PPM)



The relative position of the pulse is proportional to the amplitude of modulating signal.

Bandwidth of transmission channel depends on rising time of the pulse.

The instantaneous power of the transmitter remains constant.

Noise, interference is minimum

Simple to implement similar to phase modulation.